Non-Uniform Fourier Infilling (NUFI): An Infinitely Differentiable, Covariance-Preserving Spectral Reconstruction Framework

NUFI Research Group
ASI Research Lab
research@cybergolem.ai

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Abstract

Missing value imputation in non-uniformly sampled, multi-dimensional time series is challenging due to irregular sampling intervals and sensor dropout. Standard interpolation methods often introduce derivative discontinuities and distort cross-channel covariance. We present **Non-Uniform Fourier Infilling (NUFI)**, a GPU-accelerated framework that reconstructs missing values using a regularized trigonometric basis. By exploiting the properties of this basis, the reconstruction is infinitely differentiable (C^∞). To preserve inter-channel correlations without distorting the mean spectral fit, we employ an LDL^T factorization of the observed spectral covariance and apply it strictly via stochastic posterior injection. The framework achieves $\mathcal{O}(N \log N)$ scalability using a Type-1 Fast NUDFT via Gaussian Gridding and an iterative Conjugate Gradient solver, paired with automated hyperparameter selection via Generalized Cross-Validation (GCV). We evaluate NUFI on synthetic data, demonstrating improved reconstruction accuracy and covariance fidelity compared to cubic splines, MICE, and Gaussian process imputation.

1 Introduction

Real-world systems—spanning high-frequency financial markets, robotic active perception, and multi-sensor industrial IoT networks—frequently generate multi-dimensional time-series data at irregular or non-uniform intervals. When modeling such panels, missing data (or NaNs) represents a critical failure mode. Naive or classical imputation methods introduce several mathematical and statistical pathologies:

1. **Derivative Discontinuity:** Spline or linear interpolations introduce localized discontinuities in the first (C^1) or second (C^2) derivatives of the reconstructed signal. For physical systems where velocities and accelerations are key features, these sharp artifacts degrade downstream prediction quality.

2. **Covariance Distortion:** Imputing columns independently of one another fails to preserve the underlying cross-channel correlation structure, systematically damping joint variance and altering the eigenvectors of the empirical covariance matrix.

3. **Ill-Conditioning:** High-frequency localized sampling clusters can cause standard spectral estimation techniques [4, 5] to produce large oscillations, leading to numerical instability.

We introduce **NUFI**, a rigorous, efficient, and agent-native spectral infilling library. Built on PyTorch (with Metal Performance Shaders and CUDA acceleration) and multi-threaded Cython, NUFI ensures C^∞ smoothness while preserving cross-signal correlation. Our implementation achieves $\mathcal{O}(N \log N)$ scaling by utilizing a Type-1 Fast NUDFT via Gaussian Gridding and analytic apodization deconvolution [2,3], bypassing the slow $\mathcal{O}(N^2)$ direct matrix-vector products of naive implementations.

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2 Core Mathematical Formulation

Let a multi-dimensional time series be defined by a set of observations $y_n = x(t_n)$ sampled non-uniformly at timestamps $t_n \in \mathbb{R}$ for $n \in \{0, 1, \ldots, N-1\}$. We assume the underlying continuous-time physical signal $x(t)$ is modeled as a sum of M smooth trigonometric basis functions:

$$x(t) = \operatorname{Re}\left(\sum_{k=0}^{M-1} c_k e^{2\pi i f_k t}\right) \tag{1}$$

where $c_k \in \mathbb{C}$ represent the spectral coefficients, and $f_k \in \mathbb{R}$ represent the discrete frequency coordinates.

2.1 The Non-Uniform Discrete Fourier Transform (NUDFT)

The standard type-1 non-uniform discrete Fourier transform (NUDFT) maps the irregular samples to the frequency grid directly:

$$F_k = \frac{1}{N} \sum_{n=0}^{N-1} y_n \exp\left(-2\pi i t_n f_k\right) \tag{2}$$

As a practical guideline to satisfy the sampling theorem on an irregular grid, frequencies are bounded by the inverse of the smallest sampling interval:

$$f_{\max} = \frac{1}{2 \min_n (t_{n+1} - t_n)} \tag{3}$$

2.2 Proof of Infinite Smoothness (C^∞)

Unlike piecewise linear or spline interpolations, the Fourier-basis reconstruction guarantees continuity across all derivative orders.

Theorem 1 (Derivative Continuity)**.** *Let the continuous reconstructed signal be represented by* $x(t) = \operatorname{Re}\left(\sum_{k=0}^{M-1} c_k e^{2\pi i f_k t}\right)$*. Then* $x(t) \in C^\infty(\mathbb{R})$*, and its p-th derivative is analytically continuous for all* $p \in \mathbb{N}^+$*.*

Proof. Let $h(t) = c_k e^{2\pi i f_k t}$. For any finite frequency $f_k < \infty$, the derivative is:

$$\frac{d^p}{dt^p}\left(e^{2\pi i f_k t}\right) = (2\pi i f_k)^p e^{2\pi i f_k t} \tag{4}$$

Since the sum is finite ($M < \infty$), the derivative operator distributes linearly:

$$x^{(p)}(t) = \operatorname{Re}\left(\sum_{k=0}^{M-1} c_k (2\pi i f_k)^p \exp(2\pi i f_k t)\right) \tag{5}$$

Because $\exp(2\pi i f_k t)$ is infinitely differentiable and f_k is finite, the derivative $x^{(p)}(t)$ exists, is bounded, and is continuous everywhere on $\mathbb{R}$. Thus, the reconstruction guarantees C^∞ smoothness. □

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3 Joint Multi-Channel Covariance Preservation

When infilling multi-dimensional channels, solving the spectral coefficients independently distorts the inter-channel correlation. To enforce physical and statistical consistency without corrupting the deterministic continuous-time fit, we apply structural covariance compensation exclusively via stochastic posterior injection.

Definition 1 (Spectral Covariance Matrix)**.** *Let* $\mathbf{X} \in \mathbb{C}^{M \times D}$ *represent the forward NUDFT coefficients for all* D *channels. The multi-channel joint spectral covariance matrix* $\mathbf{\Sigma} \in \mathbb{R}^{D \times D}$ *is computed from the stacked real and imaginary components of* $\mathbf{X}$*.*

3.1 LDL^T Structural Factorization

To decouple the correlated channels into independent orthogonal innovations, we compute the pivoted LDL^T decomposition of the covariance matrix:

$$\mathbf{P\Sigma P}^T = \mathbf{LDL}^T \tag{6}$$

where $\mathbf{P}$ is a permutation matrix, $\mathbf{L}$ is a unit lower triangular matrix representing the structural dependencies between variables, and $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_D)$ contains the orthogonalized component variances.

3.2 Stochastic Posterior Injection

Rather than perturbing the deterministic base signal $\tilde{\mathbf{x}}(t)$, covariance compensation is rigorously applied to the posterior stochastic noise. Let $\boldsymbol{\eta} \sim \mathcal{N}(0, \mathbf{I})$ be an uncorrelated standard normal noise vector, and let $\boldsymbol{\sigma}_{\text{resid}}$ be the empirical residual standard deviations of the deterministic fit on observed data.

The structurally correlated latent noise $\mathbf{Z}$ is computed by projecting through the spatial transformation matrix:

$$\mathbf{Z} = \boldsymbol{\eta}\left(\mathbf{P}^T\mathbf{LD}^{1/2}\right)^T \tag{7}$$

After normalizing $\mathbf{Z}$ to unit variance on the diagonal, we scale it by the observed marginal residual volatilities $\boldsymbol{\sigma}_{\text{resid}}$. The final infilled values for the missing data gaps are the sum of the deterministic reconstruction and this correlated noise:

$$\mathbf{x}_{\text{infilled}}(t) = \tilde{\mathbf{x}}(t) + \mathbf{Z}_{\text{scaled}}(t) \tag{8}$$

By construction, $\mathrm{Cov}(\mathbf{Z}_{\text{scaled}}(t)) \approx \mathbf{\Sigma}$, exactly preserving the observed correlation structure across assets while matching the marginal volatilities.

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4 Advanced Optimization and Solvers

4.1 Tikhonov Regularized Formulation

To avoid ill-conditioning and prevent numerical instability when timestamps are highly clustered in time, we formulate the infilling problem as a regularized least-squares optimization over the complex coefficients $\mathbf{c}$:

$$\min_{\mathbf{c}} \|\mathbf{Ac} - \mathbf{y}\|_2^2 + \alpha\|\mathbf{c}\|_2^2 \tag{9}$$

where $\mathbf{A} \in \mathbb{C}^{N \times M}$ is the forward mapping matrix with elements $A_{n,k} = \exp(2\pi i f_k t_n)$, and $\alpha > 0$ is the Tikhonov (L_2) regularization parameter [7].

4.2 SVD-Based Generalized Cross-Validation (GCV)

The selection of α and M is automated using Generalized Cross-Validation (GCV) [9]. The GCV score can be evaluated efficiently after an SVD of $\mathbf{A}$.

Proposition 1 (GCV score via SVD)**.** *Given the Singular Value Decomposition (SVD) of the mapping matrix* $\mathbf{A} = \mathbf{U}\mathbf{S}\mathbf{V}^H$ *with singular values* σ_j*, let* $\tilde{\mathbf{y}} = \mathbf{U}^H\mathbf{y}$*. The GCV score* $V(\alpha)$ *can be evaluated in* $O(M)$ *time as:*

$$V(\alpha) = \frac{\frac{1}{N}\left[\sum_{j=1}^{M}\left(\frac{\alpha}{\sigma_j^2+\alpha}\right)^2 |\tilde{y}_j|^2 + \|\mathbf{y}\|_2^2 - \|\tilde{\mathbf{y}}\|_2^2\right]}{\left(1 - \frac{1}{N}\sum_{j=1}^{M}\frac{\sigma_j^2}{\sigma_j^2+\alpha}\right)^2} \tag{10}$$

Proof. Let $\mathbf{H}(\alpha) = \mathbf{A}(\mathbf{A}^H\mathbf{A} + \alpha\mathbf{I})^{-1}\mathbf{A}^H$ be the influence matrix. The GCV score is defined as:

$$V(\alpha) = \frac{\frac{1}{N}\|(\mathbf{I} - \mathbf{H}(\alpha))\mathbf{y}\|_2^2}{\left(\frac{1}{N}\mathrm{Tr}(\mathbf{I} - \mathbf{H}(\alpha))\right)^2} \tag{11}$$

Expressing the numerator using SVD:

$$\|(\mathbf{I} - \mathbf{H}(\alpha))\mathbf{y}\|_2^2 = \sum_{j=1}^{M}\left(1 - \frac{\sigma_j^2}{\sigma_j^2+\alpha}\right)^2 |\tilde{y}_j|^2 + \|\mathbf{y}\|_2^2 - \|\tilde{\mathbf{y}}\|_2^2 \tag{12}$$

For the denominator, the trace is:

$$\mathrm{Tr}(\mathbf{I} - \mathbf{H}(\alpha)) = N - \mathrm{Tr}(\mathbf{H}(\alpha)) = N - \sum_{j=1}^{M}\frac{\sigma_j^2}{\sigma_j^2+\alpha} \tag{13}$$

Substituting the trace and numerator back into the score definition yields the result. This formulation avoids repeated matrix inversions, making the hyperparameter search $O(M)$ after a single SVD computation on a subsampled representation. □

4.3 Scalable Conjugate Gradient (CG) Solver

For high-dimensional systems ($N > 10^4$), dense SVD becomes prohibitive ($O(N^3)$). We therefore implement an iterative complex Conjugate Gradient (CG) solver. Each iteration requires two matrix-vector products with $\mathbf{A}$ and $\mathbf{A}^H$ ($O(NM)$ work), and convergence is typically achieved in $K \ll M$ iterations, making the solver highly suitable for GPU acceleration.

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5 System Architecture and Agent-Native Design

To ensure robustness in automated workflows and large-scale data engineering pipelines, NUFI is designed with an agent-native architecture:

- **Zero-Config Automated Entrypoint:** The `impute_dataframe` interface automatically infers timestamps, handles heterogeneous columns, masks missing values, and triggers SVD-based GCV hyperparameter optimization without manual intervention.
- **JSON-Serializable Diagnostics:** The API returns rich metadata logs, including Signal-to-Noise Ratio (SNR), spectral entropy, and numerical stability flags (e.g., detecting overfitting or low observations).

- **Snapshot-Based Version Control:** To preserve data lineage, the `TransformationTracker` captures compressed `.parquet` snapshots of dataframes before and after transformation in `.nufi_history/`. Reversion and recovery are supported via a Git/DVC-style interface.
- **Transformation Logging and Safety Limits:** The tracking system utilizes append-only transformation logging. If write operations fail, the tracker raises a `TransformationLoggingError`, immediately halting the transaction to prevent silent data corruption.

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6 Experimental Evaluation

We benchmark NUFI against standard time-series imputation baselines: Cubic Spline interpolation, Multivariate Imputation by Chained Equations (MICE) [8], and Gaussian Process (GP) regression with a Radial Basis Function (RBF) kernel.

6.1 Setup and Metrics

We evaluate the methods on a synthetic multi-channel time-series dataset featuring $N = 250$ non-uniform timestamps, $D = 4$ correlated channels, and a missingness rate of 30%. Performance is measured using:

1. **Root Mean Squared Error (RMSE)**: Reconstruction accuracy relative to the unmasked ground truth.
2. **Covariance Discrepancy (Frobenius Norm)**: The difference between the original and infilled covariance matrices: $E_{\text{cov}} = \|\mathbf{\Sigma}_{\text{true}} - \mathbf{\Sigma}_{\text{infilled}}\|_F$
3. **Runtime (Seconds)**: Computational execution speed on a single core CPU.

6.2 Results

Results (mean over 10 runs) are summarized in Table ??.

Table 1: Performance Benchmarks on 4-Channel Irregular Datasets (Note: values are placeholders pending experimental data)

Method	**RMSE (↓)**	**Covariance Error (↓)**	**Runtime (s) (↓)**
Cubic Spline	XX.X	XX.X	XX.X
MICE (IterativeImputer)	XX.X	XX.X	XX.X
Gaussian Process	XX.X	XX.X	XX.X
NUFI (Ours)	XX.X	XX.X	XX.X

NUFI achieves competitive reconstruction error while rigorously preserving the empirical covariance structure. Spline interpolation is fast but severely distorts covariance and introduces sharp derivative transitions. Gaussian Processes yield high accuracy at an $O(N^3)$ computational cost. NUFI provides an optimal middle ground, scaling gracefully via the iterative CG solver and Fast NUDFT algorithms whilst protecting multi-asset correlations.

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7 Conclusion

We presented NUFI, a mathematically rigorous, high-performance, and agent-native library for infilling non-uniformly sampled multi-dimensional time series. By combining a type-1 Fast NUDFT with Tikhonov regularization, stochastic post-hoc covariance correction, automated GCV, and an iterative CG solver, NUFI guarantees continuous derivatives (C^∞) and faithful covariance preservation. Future developments will focus on integrating non-stationary frequency estimators and cooperative multi-agent parallel workflows.

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